

List of formulas

- Equilibrium equations:

$$\sum F_x = 0, \sum F_y = 0, \sum F_z = 0, \text{ and } \sum_{\text{about any point}} \text{Moment} = 0$$

- Statically determinacy:

$m + r = 2j \rightarrow$ Statically determinate

$m + r < 2j \rightarrow$ Mechanism

$m + r > 2j \rightarrow$ Statically indeterminate

- Stress $\sigma = \text{Force/Area}$, Strain $\varepsilon = \frac{\delta}{L}$
- Modulus of elasticity $E = \frac{\sigma}{\varepsilon}$
- The change in length, δ , due to a change in temperature, ΔT , is given by: $\delta = \alpha \cdot L_0 \cdot \Delta T$
- Centre of Area $\bar{y}$

$$\bar{y} = \frac{\int_A y \, dA}{A}$$

- Second moment of area I

$$I = \int_A y^2 \, dA$$

For rectangle $\bar{y} = 0$ and $I = \frac{bd^3}{12}$

For triangle $\bar{y} = \frac{2d}{3}$ and $I = \frac{bd^3}{4}$

Parallel axis theorem

$$I_{uu} = I_{xx} + Ae^2$$

- Relationship between bending moment, stress and curvature

$$\frac{M}{I} = \frac{\sigma}{y} = \frac{E}{R}$$

- Deflection of beams

$$EI \frac{d^2 v}{dx^2} = -M$$

- Stress due to shear force

$$\tau = F \frac{A\bar{y}}{Ib}$$

$$\text{Shear flow } q = \frac{FA\bar{y}}{I}$$